WEEK 2 | DATASET AUDIT REPORT

Executive summary

The supplied dataset contains 6,500 tertiary-education reviews: 3,000 teacher reviews, 3,000 course reviews and 500 university reviews. The only actual null values are 45 missing numerical ratings (0.69%); no review text is missing. There are no duplicate full records or IDs, although nine review-text entries repeat the text of earlier entries. Rating-based categories are skewed towards Pos (60.02% of all records). Six descriptive distribution figures and one additional rating histogram are provided. A separate qualitative sample of 60 reviews demonstrates that mixed, context-dependent and ironic wording complicates a simple positive/negative/neutral classification.

1. Dataset description

EduRABSA is the dataset selected for in the internship instructions. The supplied CSV is a review-level extract with metadata, numerical ratings, review text and rating-derived category fields. Its three review types cover teaching staff, courses and universities. The internship instructions describe a broader dataset with manually annotated aspect-based sentiment information, those detailed annotations are not columns in this CSV. Consequently, review-level descriptive results below refer only to the uploaded extract, not to unprovided aspect labels.

The raw dataset was read without deleting or modifying rows. Derived character and word counts were calculated separately. For numerical rating analysis, missing entries were excluded only from summaries requiring a rating.

2. Data dictionary

```
Column names:
['type', 'source', 'id', 'rating', 'num_token', 'sample_criteria', 'review_text', 'rating_bin', 'rating_cat', 'length_bin']

Dataset information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6500 entries, 0 to 6499
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   type             6500 non-null   object
 1   source           6500 non-null   object
 2   id               6500 non-null   object
 3   rating           6455 non-null   float64
 4   num_token        6500 non-null   int64
 5   sample_criteria  6500 non-null   object
 6   review_text      6500 non-null   object
 7   rating_bin       6500 non-null   object
 8   rating_cat       6500 non-null   object
 9   length_bin       6500 non-null   object
```

Operational note: [NA] in rating_cat and rating_bin is a literal category string, not a pandas null. Conf/Neu combines conflicting/neutral rating-derived entries and must not be represented as a verified neutral text label. The meaning of numeric ratings differs between sources (for example, course-source value 1.0 is grouped as Pos), so a universal numerical cutoff is inappropriate without source-specific documentation.

3. Sample size and review composition

The audit covers 6,500 records and 10 original columns. There are 6,500 unique IDs. Every review has a non-missing text field.

type	Frequency	Percentage
teacher review	3000	46.15
course review	3000	46.15
university review	500	7.69

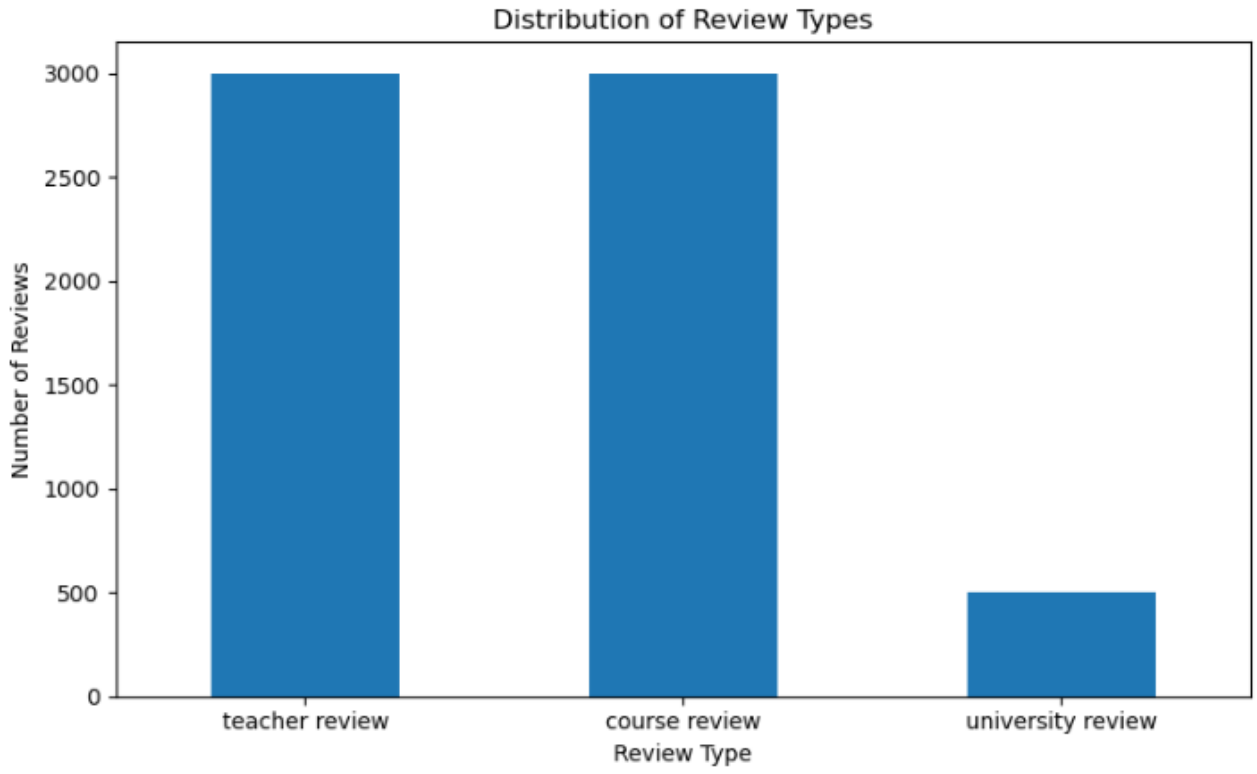


Figure 1. Count of each review type.

4. Missing values

	Missing_Count	Missing_Percentage
type	0	0.00
source	0	0.00
id	0	0.00
rating	45	0.69
num_token	0	0.00
sample_criteria	0	0.00
review_text	0	0.00
rating_bin	0	0.00
rating_cat	0	0.00
length_bin	0	0.00

Only rating has actual missing values: 45 of 6,500 (0.69%), leaving 6,455 recorded ratings. The corresponding rating_cat and rating_bin fields use the literal marker [NA] for these same 45 cases. No empty-string values were observed in the object/text fields after trimming whitespace.

5. Duplicates

```
Duplicate rows: 0
Duplicate IDs: 0
Duplicate review texts: 9
```

The nine repeated text entries are not nine identical full records: they have distinct review IDs and are predominantly short generic expressions (for example, “Great class!” and “Fun course”). They should be flagged for review rather than indiscriminately deleted. Text repetition can be genuine across different reviews.

6. Label distribution and class imbalance

rating_cat	Frequency	Percentage
Pos	3901	60.02
Neg	1731	26.63
Conf/Neu	823	12.66
[NA]	45	0.69

Among the 6,455 non-[NA] rating-category entries, Pos is the majority class; the ratio of Pos to Neg is 2.25:1, and Pos to Conf/Neu is 4.74:1. This is a descriptive imbalance that should inform later stratification and evaluation. These categories are rating-derived; the audit does not claim they are the final manually annotated text-sentiment labels.

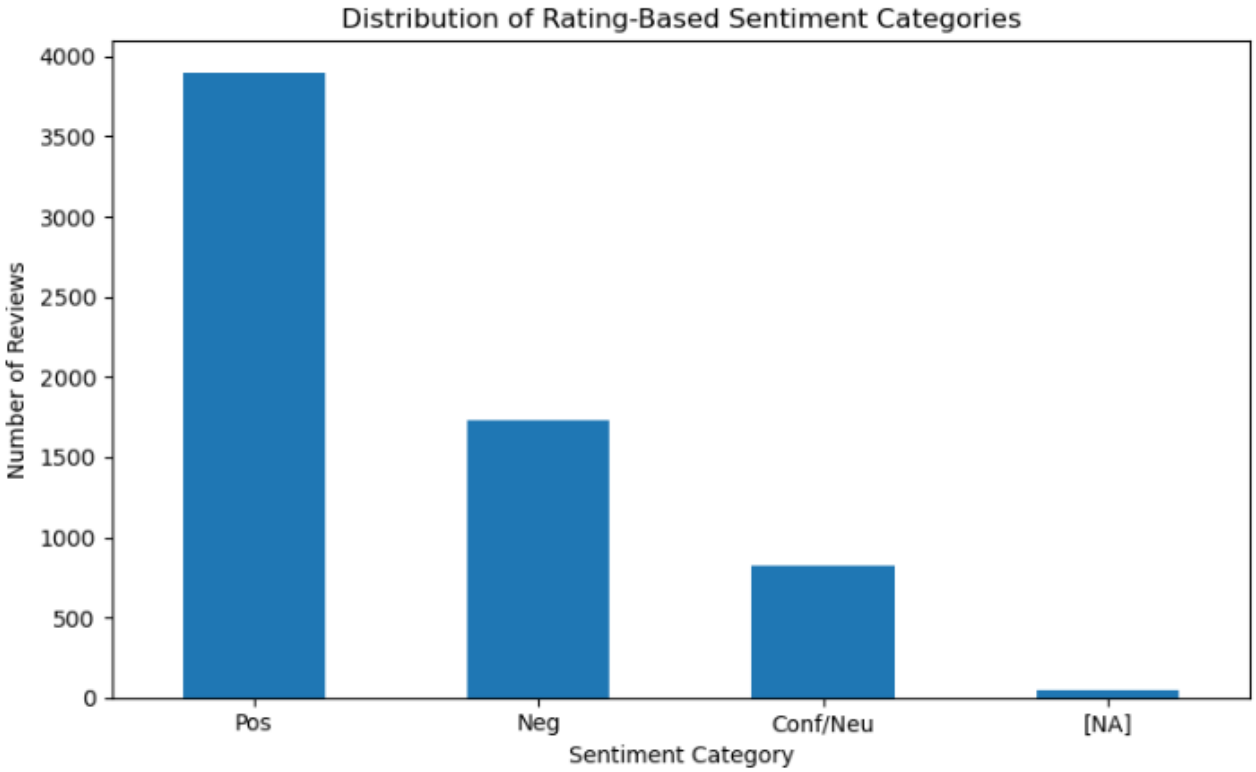


Figure 2. Frequency of original rating-derived categories.

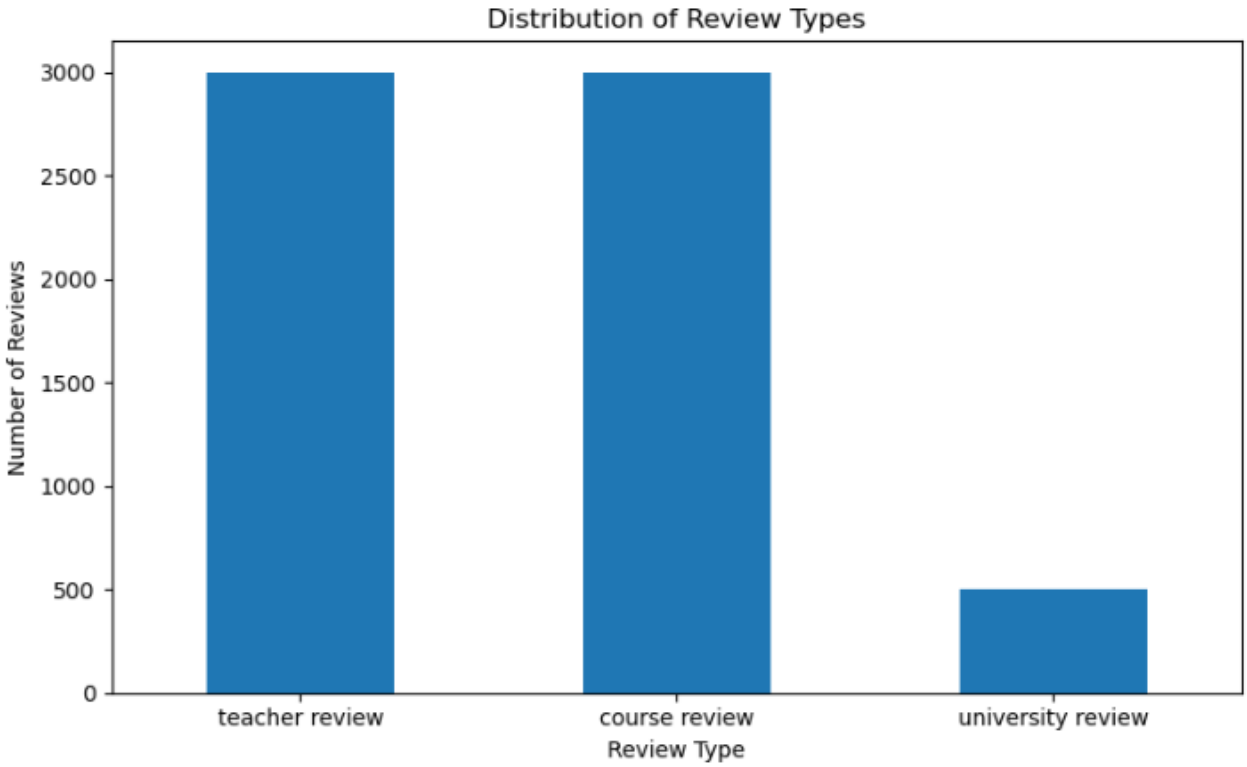


Figure 3. Original rating-derived categories across teacher, course and university reviews.

7. Text-length summaries

	character_count	word_count	num_token
count	6500.00	6500.00	6500.00
mean	220.94	39.63	45.29
std	182.34	32.71	37.09
min	2.00	1.00	1.00
25%	95.00	17.00	19.00
50%	192.00	34.00	40.00
75%	315.00	56.25	64.00
max	2508.00	437.00	512.00

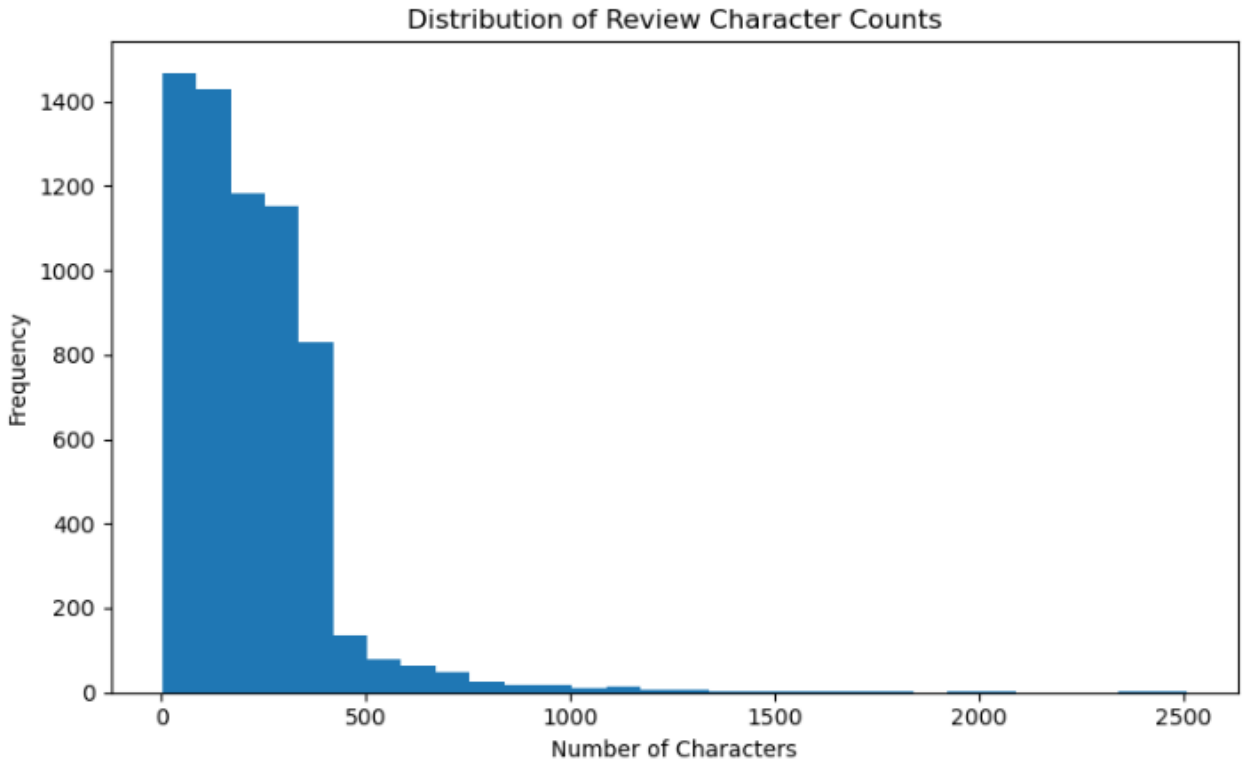


Figure 4. Distribution of characters per review.

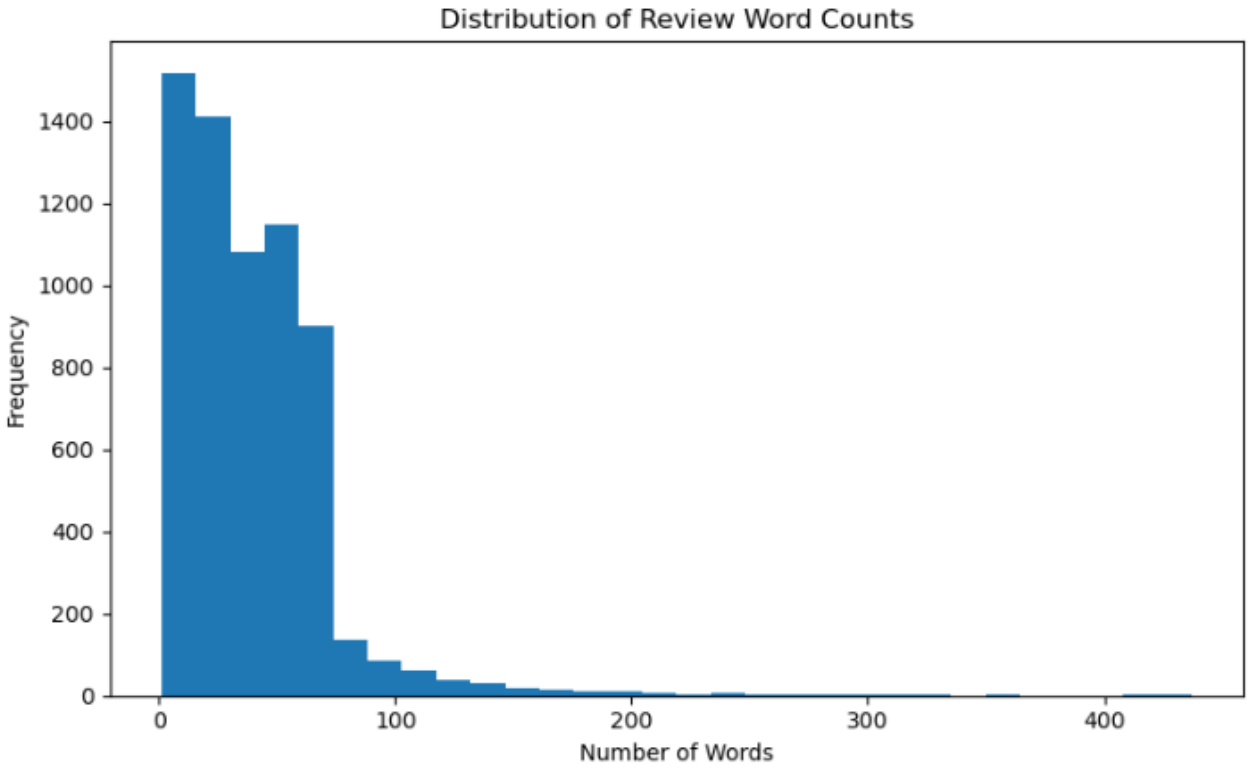


Figure 5. Distribution of whitespace-separated words per review.

8. Figures and aspect-category availability

Figures 1–5 document overall category frequencies, review-type frequencies, categories by review type, character count, word count and recorded token count. Figure 7 below supplements these with the numerical rating distribution (n = 6,455 non-missing ratings). At least five useful figures are therefore present in this audit.

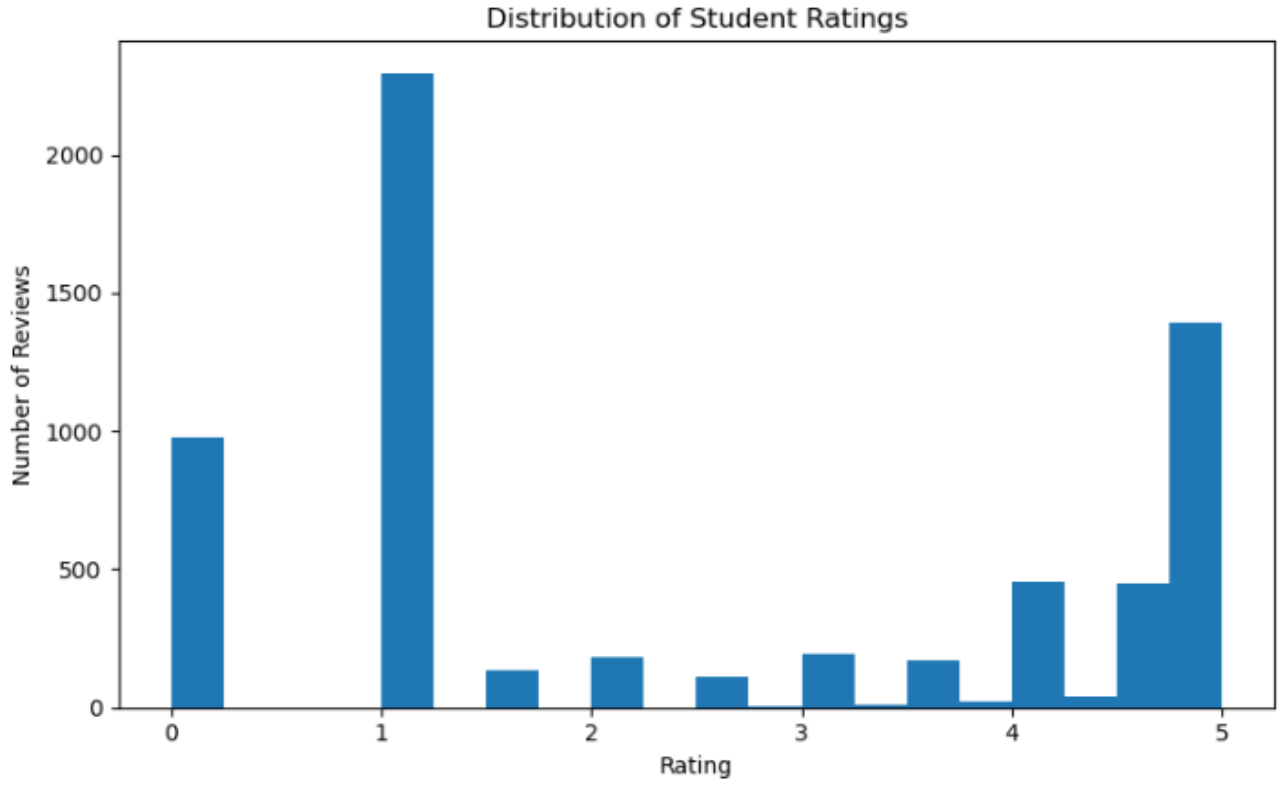


Figure 7. Distribution of numerical rating values, excluding 45 missing ratings.

9. Manual inspection

Manual text sentiment	Count
Positive	22
Negative	17
Mixed	19
Neutral	1
Uncertain	1
Total	60

Primary observation type	Count
Clear	31
Mixed	17
Context-dependent	8
Ambiguous	2
Sarcastic	2
Total	60

The following examples are excerpted from the full reviews the accompanying interpretation uses the complete text in the annotated workbook.

	id	rating_cat	rating	review_text	manual_sentiment	observation_type
2	1_83_2470_12673	Pos	5	I highly recommend taking this class if you are a freshmen. The lectures are once a week and you are required to observe at a school site for 45 hours. I did 4 hours a week and it was very simple. Mrs. Schlievert is a very easy going professor she had lectures that were conversational. We always got out early and I learned a lot. Overall easy A	Positive	Clear
3	2_10110_1185	Pos	1	Say good bye to your grades for this course! I feel Chong makes this course harder than it should be his slides/annotation were all over the place. I had to more often than not teach myself the course through websites and the text book. That being said despite being taught by Chong I liked the course a lot.	Mixed	Mixed
4	2_3107_168	Pos	1	If you did well in Calc 1, you'll do well in Calc 2	Positive	Context-dependent
	3_171_510_271	Pos	4.4	Great uni, nice societies , good lecturers	Positive	Clear

10. Potential limitations of the dataset

- **Missing numerical ratings:** 45 entries lack recorded numeric ratings and are marked [NA] in derived rating categories.
- **Rating-based labels versus text sentiment:** Some reviews express multiple opinions; source-specific rating encodings and category derivation mean rating_cat should not be assumed to equal manually annotated overall sentiment.
- **Imbalanced original categories:** Pos accounts for about 60% of all records; Conf/Neu is considerably smaller and absent from the course-source extract.
- **Unequal review-type representation:** Teacher and course reviews each number 3,000, while university reviews number 500.
- **Repeated brief review text:** Nine repeated text entries beyond first occurrence can affect simple text-frequency analysis; IDs remain unique.

- **Length variation and short reviews:** Review length ranges from 1 to 437 whitespace-separated words; extremely short reviews may lack interpretive context.
- **Aspect information not present in supplied extract:** The annotated ABSA file is required before reporting aspect-category distribution, aspect-level missingness or annotation quality.
- **Generalisation and qualitative subjectivity:** English tertiary education reviews may not generalise directly to multilingual Malaysian or medical education settings; the 60-review manual coding was conducted by one reviewer.